Beginner Instruction Note

Starting a Materials Screening Study for Ultra Thin Hand and Finger Protection in Synchronized Figure Skating

A practical preliminary note for researchers beginning a larger materials and wearable design project

Purpose of this note

This note is meant to help a beginner begin confidently. The present topic is highly practical and absolutely researchable. The aim of the preliminary phase is not to solve the entire problem at once. Rather, it is to create a strong and defensible starting point by doing three things well:

1. understand the injury mechanism and why this problem is unique,
2. become familiar with the most relevant families of protective materials, and
3. screen a manageable set of candidate materials using an accessible database before moving to prototype design and experiments.

The project is motivated by a very specific unmet need: synchronized skaters perform in close formation, and hand and finger injuries can occur when a fallen skater's exposed hand is contacted by the skate blade of a following skater. Existing common handwear solutions do not solve this problem because they are either too bulky, not cut resistant, or not designed for simultaneous cut protection, impact attenuation, and dexterity.

1. First, understand the problem as a materials problem

A beginner should translate the sporting problem into a materials design problem. The wearable must satisfy several requirements at the same time:

- it must be **thin**, so that grip, hand holding, and choreography are not disrupted,
- it must provide **cut resistance**, especially against sharp blade contact,
- it should provide at least some **impact attenuation**, because falls on ice involve local impact as well,
- it must preserve **range of motion and tactile control**,
- it should remain functional at **low temperature**, and

- it should ideally be manufacturable as **finger sleeves, back-of-hand inserts, or a hybrid glove architecture**.

This immediately tells us that one single material may not be enough. A layered or hybrid solution is likely to be more realistic than a monolithic one.

2. Five articles the researcher should know at the beginning

No paper appears to directly solve the exact problem of ultra thin hand and finger protection for synchronized figure skating. That is precisely why this topic is promising. However, a strong starting point can still be built by combining insights from adjacent literature. The following papers should be read first.

2.1. 1. Dubravcic-Simunjak et al. — *Injuries in Synchronized Skating*

Why read it: This is one of the most important sport-specific starting papers because it focuses on synchronized skating rather than skating in general.

What it offers: The paper documents injury patterns in synchronized skaters and establishes that the discipline has its own injury profile. Even if the paper does not present a materials solution, it helps a beginner justify why synchronized skating should be treated as a distinct safety problem rather than merely a subset of general skating injury literature.

What to learn from it: Use this paper to frame the sporting context, team formation risk, and the need for discipline-specific protective design.

Reference link: https://pubmed.ncbi.nlm.nih.gov/16767614/

2.2. 2. Murphy et al. — *Ice-skating Injuries to the Hand*

Why read it: This paper is especially valuable because it directly highlights hand injuries in skating and explicitly points toward protective gloves as a preventive measure.

What it offers: The paper reports that hand lacerations, especially over the dorsum of the hand, are common in skating injuries. This is extremely important for your project because it validates that the hand is not a peripheral issue but a genuine protection gap.

What to learn from it: This paper supports the argument that hand protection is medically meaningful. It can be used to justify why a wearable should protect not only the palm but also exposed fingers and dorsal surfaces.

Reference link: https://pubmed.ncbi.nlm.nih.gov/2230504/

2.3. 3. Zhai et al. — *Research Progress of Cut Resistant Textile Materials*

Why read it: This is the most useful beginner review for understanding the material side of cut resistance.

What it offers: The paper discusses commonly used cut resistant materials such as ultra high molecular weight polyethylene, aramids, glass fiber blends, and coating strategies. It also explains standard test approaches such as EN 388, ASTM F1790, and ISO 13997. Importantly, it emphasizes that future development should pursue lightness, softness, and comfort in addition to protection.

What to learn from it: This paper helps the researcher move from vague phrases such as "strong material" to meaningful families such as UHMWPE based fibers, para aramids, hybrid yarns, coated fabrics, and flexible composite systems.

Reference link: https://pmc.ncbi.nlm.nih.gov/articles/PMC8511638/

2.4. 4. Parisi et al. — *Use of Shear Thickening Fluids in Sport Protection Applications*

Why read it: This paper opens the door to an important design idea for the present topic: materials that remain flexible during normal use but stiffen or dissipate energy more effectively under sudden impact.

What it offers: The review summarizes the use of shear thickening fluid based systems in sport protection and discusses their promise relative to bulky traditional pads. Although the current skating problem is not identical to helmet or body padding, the underlying idea of flexible protective response is highly relevant.

What to learn from it: This paper encourages the beginner to think beyond ordinary foam padding. It suggests that a hybrid system combining a cut resistant textile with a thin impact responsive layer may be a fruitful direction.

Reference link: https://www.frontiersin.org/journals/materials/articles/10.3389/fmats.2023.1285995/full

2.5. 5. Khanlari et al. — *Protective Gloves, Hand Grip Strength, and Dexterity Tests*

Why read it: This paper is essential because the present project is not only about protection. It is equally about preserving function.

What it offers: The paper compares protective gloves using dexterity and grip related tests and shows that glove thickness and construction can strongly affect hand performance.

What to learn from it: This paper will remind the beginner that a "better protected" glove is not automatically a "better design." If dexterity becomes poor, synchronized skaters may reject the design. Therefore, hand function must be treated as a design variable from the very beginning.

Reference link: https://pmc.ncbi.nlm.nih.gov/articles/PMC9947274/

3. How these five papers fit together

Taken together, these five papers give a very usable first framework:

- the **skating injury papers** tell you why the problem exists,
- the **cut resistant textile review** tells you which material families matter,
- the **sport impact review** expands the design space toward thin adaptive protection, and
- the **dexterity paper** keeps the design grounded in actual user function.

That combination is more powerful than reading only glove papers or only sports injury papers in isolation.

4. The suggested starting database: MatWeb

4.1. Database link

https://www.matweb.com/

4.2. What MatWeb is

MatWeb is an online materials information resource that contains a large searchable collection of engineering material datasheets, including polymers, fibers, metals, ceramics, and composites. For a beginner, its biggest advantage is that it is structured like a practical engineering lookup system rather than a highly specialized research database. It is therefore a good first place to compare commercially relevant material families.

4.3. How it is built and why it is useful

MatWeb is built as a property database composed of material datasheets supplied and organized across many engineering material classes. The database supports searching by material category, text keywords, manufacturer information, and property values. It also offers property-based search tools where users can filter materials using ranges for quantities such as density, tensile strength, modulus, elongation, impact related measures, tear strength, and service temperature.

For the present problem, this matters because the early phase of the work is not yet about proving final performance. It is about reducing a huge universe of possible materials into a small shortlist that is worth deeper study.

4.4. Why this database is a good first step for this topic

MatWeb is appropriate at the preliminary stage for three reasons:

1. it is **accessible to beginners**,
2. it is broad enough to let you compare **soft polymers, elastomers, films, fibers, and composites**, and
3. it allows you to move from intuition to a **property guided shortlist**.

A very important caution should be noted. MatWeb will not directly tell you whether a material is safe against a figure skate blade. It is a screening tool, not the final proof. The final case must later be built using literature, standards, and experiments. That is perfectly acceptable. In fact, this is how many strong materials projects begin.

5. How a beginner should screen the database

5.1. Step 1: Do not search for a final material immediately

Do not begin by asking, "What is the best glove material?" That usually leads to confusion. Begin instead by searching for **material families**. For this topic, a practical first set is:

- thermoplastic polyurethane,
- polyurethane elastomers,
- thermoplastic elastomers,
- nylon based films or fibers,
- ultra high molecular weight polyethylene,
- aramid related materials,
- thin composite sheets or laminates.

These are not all final candidates. They are simply reasonable starting families because they represent different balances of flexibility, toughness, cut resistance potential, and manufacturability.

5.2. Step 2: Translate the design need into screening criteria

The beginner should create a small table before opening the database. An example is shown below.

Design need	What to look for in early screening
Thinness	film or sheet forms, fiber forms, coatings, lightweight materials
Flexibility	moderate or low modulus, high elongation, elastomeric behavior
Cut resistance potential	tough fibers, high strength polymers, protective textile relevance
Impact attenuation	impact related properties, tough elastomers, viscoelastic systems
Cold use	acceptable minimum service temperature, no brittle behavior near ice rink conditions
Wearability	low density, softness, flexible architecture potential

This table helps the student search intentionally instead of randomly.

5.3. Step 3: Use beginner friendly search strategies in MatWeb

A beginner can use MatWeb in three simple modes.

Mode A: Text search by material family. Start with broad words such as "TPU," "UHMWPE," "aramid," "polyurethane elastomer," or "thermoplastic elastomer." Open a few datasheets and simply learn how the information is presented.

Mode B: Category based browsing. Browse through plastics, elastomers, fibers, or composite categories. This is helpful when you do not yet know exact trade names.

Mode C: Property based filtering. Once you gain confidence, use MatWeb's property search to filter by density, elongation, tensile strength, tear strength, modulus, and service temperature. This is where the shortlist begins to emerge.

5.4. Step 4: Begin with elimination, not perfection

A beginner does not need to identify the best material in the first week. Instead, eliminate obviously unsuitable classes.

For example:

- materials that are too stiff are likely to hurt dexterity,
- materials that are very soft but weak may not help against blade contact,
- materials with poor low temperature performance are risky,

- heavy materials are unlikely to be acceptable in a sport setting.

This elimination mindset is liberating. It makes the task feel manageable.

5.5. Step 5: Build a first shortlist of 8 to 12 candidates

A good beginner target is not fifty materials. It is around eight to twelve. That is enough to reveal patterns without becoming overwhelming.

For each shortlisted material, record the following in a spreadsheet:

- material name or grade,
- family,
- density,
- tensile strength,
- elongation at break,
- modulus if available,
- tear or impact related values if available,
- minimum service temperature if available,
- form availability: film, fiber, sheet, coating, fabric,
- first impression: impact layer candidate, cut layer candidate, or hybrid layer candidate.

6. A very practical screening logic for this project

The present problem becomes much easier if the candidate materials are grouped by function rather than treated as competitors in one single list.

Group A: Soft energy absorbing layers

These are materials that may reduce local impact and improve comfort. Examples to explore include polyurethanes, thermoplastic polyurethanes, and some thermoplastic elastomers.

Group B: Cut resistant structural layers

These are materials that may help resist slicing or tearing. Examples to explore include UHMWPE based fibers or fabrics and para aramid based systems.

Group C: Hybrid concepts

These combine a thin cut resistant textile layer with a compliant backing or a thin adaptive layer such as a shear thickening system.

This functional grouping is powerful because it prevents the common beginner mistake of expecting one material to do everything perfectly.

7. How to use large language models to make the work easier

This part is important. A beginner does not need to code everything alone. Tools such as ChatGPT can help a great deal, especially in the preliminary stage.

7.1. What LLMs can help with

- converting your screening logic into a spreadsheet template,
- generating Python code to clean and organize copied datasheet values,
- suggesting plots such as density versus tensile strength or elongation versus modulus,
- helping write small scripts to rank candidates using weighted scoring,
- summarizing differences among shortlisted materials in plain language,
- helping draft a literature matrix for the five starting papers.

7.2. A beginner friendly way to use ChatGPT

You can literally ask for coding help in small steps. For example:

> "I have copied material properties from MatWeb into a CSV file. Please write beginner friendly Python code that reads the CSV, removes empty cells, and makes a ranked shortlist based on low density, high elongation, and high tensile strength."

or

> "Please create a weighted screening method for candidate glove materials where flexibility is weight 0.35, cut resistance potential is weight 0.30, impact protection potential is weight 0.20, and low temperature suitability is weight 0.15."

This makes the project feel far more doable. You are not expected to become an expert programmer before beginning the materials work.

8. A simple preliminary workflow that is absolutely doable

A beginner can complete a credible preliminary study in the following sequence.

1. Read the five papers and make a one page summary table.
2. Define the design requirements in your own words.
3. Use MatWeb to explore 5 to 7 material families.
4. Shortlist 8 to 12 candidate materials.
5. Group them into impact layer, cut layer, and hybrid candidates.
6. Use a spreadsheet and simple plots to compare them.
7. Use ChatGPT to help organize the data and generate small screening scripts.
8. End the preliminary phase by identifying 2 or 3 promising design architectures rather than one final material.

That final point is very important. A strong preliminary study may conclude with designs such as:

- a thin finger sleeve with UHMWPE outer layer and soft polyurethane inner layer,
- a dorsal hand insert using a compliant thin shell over a cut resistant textile,
- a hybrid glove concept with localized protection only at high risk regions.

That is already a meaningful research outcome for a first phase.

9. What success looks like at the end of the preliminary stage

At the end of the first stage, success does not mean a finished commercial glove. Success means:

- you understand the injury mechanism,
- you know the relevant protective material families,
- you have a rational shortlist supported by database screening,
- you can justify why certain materials were rejected,
- you have 2 or 3 plausible prototype concepts for deeper study.

If you can reach that point, you already have a solid foundation for a larger and more rigorous research project.

Closing note

This topic may look specialized, but the starting work is very approachable. You are not trying to invent the final solution on day one. You are trying to build a clear pathway from injury mechanism to candidate material families to a defendable shortlist. With a few carefully chosen papers, a beginner friendly database such as MatWeb, and the assistance of coding support from large language models, this is entirely doable.

Quick reference list

1. Dubravcic-Simunjak S, et al. *Injuries in Synchronized Skating*. https://pubmed.ncbi.nlm.nih.gov/16767614/

2. Murphy NM, et al. *Ice-skating Injuries to the Hand*. https://pubmed.ncbi.nlm.nih.gov/2230504/

3. Zhai Y, et al. *Research Progress of Cut-Resistant Textile Materials*. https://pmc.ncbi.nlm.nih.gov/articles/PMC8511638/

4. Parisi M, et al. *Use of Shear Thickening Fluids in Sport Protection Applications*. https://www.frontiersin.org/journals/materials/articles/10.3389/fmats.2023.1285995/full

5. Khanlari P, et al. *Protective Gloves, Hand Grip Strength, and Dexterity Tests*. https://pmc.ncbi.nlm.nih.gov/articles/PMC9947274/

6. MatWeb materials database. https://www.matweb.com/